

[Re] Do neutrinos dream in 5D? Partial replication of the Brane–Dirac spectrum and an audit of open exclusion reconstruction

Keiji Yoshimura¹

¹Independent Researcher

ORCID: 0009-0007-6631-1038

Correspondence: keiji.yoshimura.research@gmail.com

Submission status

Pre-submission final

Original article

de Giorgi, Pasari & Turner
(2026)

DOI

10.1007/JHEP05(2026)152

Replication class

[Re] – partial
(Figure 1 subset only)

A partial replication of A. de Giorgi, D. Pasari, and J. Turner, *Do neutrinos dream in 5D? Towards a comprehensive extra-dimensional neutrino phenomenology*, JHEP 2026, 152 (2026).

Abstract

We report a partial computational replication of the Brane–Dirac spectrum in Figure 1 of de Giorgi, Pasari, and Turner, together with a bounded audit of an attempted open reconstruction of their experiment-level exclusion contours. The successful replicated subset is restricted to the theoretical spectrum defined by Eqs. (3.26)–(3.28). Starting from the published equations alone, without using the target authors’ code, our previous LED kernel, Newtrinos, or a Daya Bay likelihood, we independently solved the infinite-tower eigenvalue equation for $m_D = 1$ and $\mu_1 = \{10, 1, 0.1\}$ and reproduced the corresponding mixing-normalisation structure. A second implementation based on finite KK sums at $N = 5000, 10000, 20000$ followed by $1/N$ extrapolation agreed with the closed-form roots at all nine preregistered anchors, with a maximum mass difference of 1.29×10^{-9} . The truncated sums of N_χ^2 exceeded the target paper’s 99% numerical criterion for all three parameter choices. An initially failed raw equation-residual gate was subsequently shown to be ill-conditioned near cotangent poles: all 3552 plotted roots lie within the requested solver-coordinate tolerance under a derivative-scaled backward-error test, with a maximum normalized error of 0.2501; two worst cases were independently confirmed at 80-digit precision. We also tested four digitized Daya Bay points from Figure 5. None produced a unique point-wise replication candidate because the open comparator crossed the $\Delta\chi^2 = 4.605$ threshold multiple times within each digitization interval. Separately, the public Daya Bay covariance-fit reference was reproduced in the frozen environment to an absolute χ^2 difference of 1.33×10^{-11} , while our earlier fixed-coordinate profiling route was shown to be non-equivalent to that public full fit. We therefore classify the work as a partial replication of Figure 1 only; no replication of the published exclusion contours, no new LED signal, and no error in the original analysis are claimed.

1 Introduction

Large-extra-dimension (LED) neutrino models introduce Kaluza–Klein (KK) towers of gauge-singlet neutrino states that modify active-neutrino oscillations. Their phenomenology and experimental constraints have been studied for more than two decades.^{1,2,3,4} Recent analyses continue to refine reactor and accelerator constraints.^{5,6}

The target of this replication is the 2026 analysis by de Giorgi, Pasari, and Turner.⁵ That article develops several five-dimensional neutrino-mass scenarios, derives their spectra and mixing patterns, and confronts them with oscillation data. Its Figure 1 presents the theoretical Brane–Dirac spectrum for $m_D = 1$ and $\mu_1 = \{10, 1, 0.1\}$; later figures present

experiment-level exclusion contours. The article's published availability statements report no deposited analysis code or associated target-analysis data.⁵

This project originally focused on reconstructing the exclusion contours from public experimental products. Reviewer-style reanalysis showed that two early interpretations were too strong: an apparent 39-versus-17 threshold-crossing difference had confounded KK truncation with unequal radius-grid resolution, and a binary Daya Bay “calibration failure” had compared a project-defined fixed-coordinate profiling route with a non-equivalent public full-fit construction. Both claims were withdrawn before the present manuscript rebuild.

The revised replication question is therefore narrower and cleaner:

Can the computational core of the published Brane–Dirac spectrum be independently reproduced from the equations in the target article, and how far can that successful theoretical replication be extended toward the experiment-level exclusion contours using open information?

The answer reported here has two parts. The Figure 1 spectrum is independently reproduced and numerically triangulated. The Figure 5 extension is not: four directly digitized Daya Bay contour points remain locally ambiguous in the tested open comparator. The partial-replication claim is consequently restricted to the theoretical Figure 1 subset.

2 Replication target and claim boundary

2.1 Target result: Brane–Dirac spectrum

For the Brane–Dirac scenario, the target article writes the finite-tower eigenvalue condition and its infinite-tower closed form as⁵

$$\sum_{n=0}^N \frac{(\chi_n m_D)^2}{m_\lambda^2 - \mu_n^2} = 1 \quad \xrightarrow{N \rightarrow \infty} \quad \pi \cot\left(\frac{\pi m_\lambda}{\mu_1}\right) = \frac{\mu_1 m_\lambda}{m_D^2}, \quad (1)$$

with $\mu_n = n\mu_1$, $\chi_0 = 1$, and $\chi_{n>0} = \sqrt{2}$. The corresponding infinite-tower normalisation is

$$N_\lambda = \left[\frac{1}{2} \left(\frac{\pi^2 m_D^2}{\mu_1^2} + \frac{m_\lambda^2}{m_D^2} + 1 \right) \right]^{-1/2}, \quad (2)$$

and unitarity implies

$$\sum_{\lambda} N_\lambda^2 = 1. \quad (3)$$

The original Figure 1 plots Eq. (2) continuously and marks the physical roots of Eq. (1) for $m_D = 1$ and $\mu_1 = 10, 1, 0.1$.⁵

2.2 What counts as replicated

The successful replication claim is limited to the computational content of Figure 1 and Eqs. (1)–(3). It does not include the Daya Bay or MINOS/MINOS+ likelihoods, the Figure 5 exclusion contours, or any target-author implementation details that are not public. The Figure 5 work is retained because it tests the boundary of the open reconstruction. It is reported as an unresolved extension, not as a failed replication of the target article.

3 Independent Figure 1 implementation

3.1 Closed-form infinite-tower route

The primary implementation solves Eq. (1) directly in the dimensionless coordinate $y = m_\lambda/\mu_1$. Each positive physical root lies on the positive-cotangent branch $n < y < n+1/2$. Brent root finding was applied independently in each branch. The normalisation N_λ was then evaluated from Eq. (2). No code from the target authors, no earlier project LED kernel, no Newtrinos component, and no experimental likelihood was imported into this calculation.

3.2 Independent finite-KK route

A second implementation evaluates the finite sum on the left-hand side of Eq. (1) directly, using $\chi_0 = 1$, $\chi_{n>0} = \sqrt{2}$, and $\mu_n = n\mu_1$. Roots were obtained at $N = 5000, 10,000$, and $20,000$. The two finest values were combined with a leading- $1/N$ Richardson extrapolation. Before the result was inspected, nine anchors were fixed: $\mu_1 = 10, 1, 0.1$ crossed with mode labels $n = 0, 1, 5$. A numerical-identity tolerance of 10^{-7} in mass was used only to compare the two independent implementations; it is not a physics acceptance tolerance.

3.3 Unitarity and asymptotic diagnostics

For each published μ_1 , the number of roots included in the sum of Eq. (3) was increased until the partial sum exceeded 0.99. The 99% level is not project-defined: the target article states that its numerical analyses include enough modes to satisfy $\sum_\lambda N_\lambda^2 > 99\%$, with the required number depending on model parameters.⁵

The large-spacing and small-spacing limits discussed around Eqs. (3.29) and (3.31) of the target article were evaluated as descriptive diagnostics. No additional pass/fail tolerance was invented for those asymptotic approximations.

4 Figure 1 replication results

4.1 Independent numerical triangulation

All nine preregistered finite-sum/infinite-form comparisons passed. Table 1 reports the largest absolute mass difference within each published μ_1 group.

Table 1. Independent numerical triangulation of Figure 1 roots. Each row summarizes modes $n = 0, 1, 5$.

μ_1/m_D	Maximum $ m_{\text{extrap}} - m_{\text{closed}} $	Status
10	2.31×10^{-11}	PASS
1	1.29×10^{-9}	PASS
0.1	1.31×10^{-10}	PASS

The unitarity check also passed for all three published parameter choices. The first truncations in the adaptive sequence to exceed 0.99 were 64, 256, and 32768 modes for $\mu_1 = 10, 1, 0.1$, respectively, giving partial sums 0.9996850, 0.9921727, and 0.9938966. The strong increase in the required mode count as μ_1 decreases is consistent with the target article's statement that many KK modes participate when the spacing is small.⁵ The independently generated spectrum is shown in Figure 1. The lightest-state overlap decreases from $N_\lambda = 0.9837$ at $\mu_1 = 10$ to 0.4249 at $\mu_1 = 1$ and 0.0450 at $\mu_1 = 0.1$. At

$\mu_1 = 10$, the large-spacing approximation differs from the exact lightest mass by approximately 3.0×10^{-4} fractionally. At $\mu_1 = 0.1$, the median of the first 50 exact N_λ values differs from the small-spacing plateau approximation by approximately 3.65×10^{-3} . These asymptotic comparisons are descriptive corroboration rather than acceptance gates.

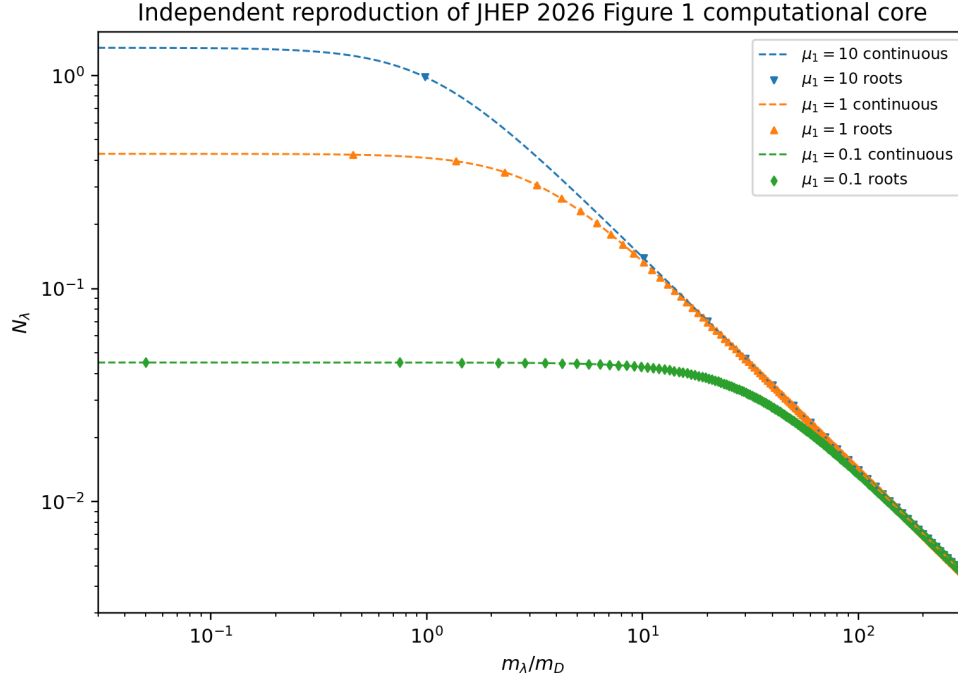


Figure 1. Independent reproduction of the computational core of the target article’s Figure 1. Dashed curves evaluate the published infinite-tower normalisation continuously; markers show independently solved physical roots for $m_D = 1$ and $\mu_1 = 10, 1, 0.1$. This figure was generated from the published equations and is not copied from the target article.

4.2 Conditioning-aware validation of all plotted roots

The first implementation included an additional raw equation-residual gate, $|F(y)| < 10^{-9}$, for every plotted root of

$$F(y) = \pi \cot(\pi y) - \left(\frac{\mu_1}{m_D} \right)^2 y. \quad (4)$$

That gate failed because the maximum raw residual was 2.67×10^{-6} . This automatic failure is preserved in the audit trail; it was not silently overwritten.

A subsequent conditioning audit showed that the raw function residual is a poor coordinate-error metric near cotangent poles, where $|F'(y)|$ is large. We therefore evaluated the first-order backward coordinate error $|F/F'|$ for all 3552 plotted roots and normalized it by the already requested Brent coordinate tolerance, $\text{xtol} + \text{rtol}|y|$. All 3552 roots lie within that requested coordinate tolerance. The maximum normalized error is 0.250095, the median is 0.002510, and the 99th percentile is 0.239561.

The root with the largest raw residual, at $\mu_1 = 10$, $n = 30$, was independently re-solved at 80-digit precision; its mass differed from the double-precision result by 2.90×10^{-12} . A second 80-digit check at the root with the largest normalized backward error differed by 2.73×10^{-12} in mass. No replacement raw-residual threshold was introduced. The original gate is therefore interpreted as a technical diagnostic failure caused by conditioning, while the Figure 1 root coordinates remain numerically validated.

5 Attempted extension to Figure 5

5.1 Direct contour digitization

The target article’s Figure 5 gives 90% confidence contours for the Dirac-brane scenario, with solid curves corresponding to Daya Bay and dotted curves to MINOS/MINOS+.⁵ Four Daya Bay points were selected for a direct test: normal and inverted ordering at $m_{\text{lightest}} = 0.3$ and 1.0 eV. The published curves were digitized from the version-of-record PDF using multiple raster resolutions and color-threshold variants. All four points passed the frozen digitization QA.

Table 2. Direct Figure 5 digitization and local open-comparator outcome. “Minimum crossings” is a certified lower bound within the digitization interval at $N = 192$.

Ordering	m_{lightest} [eV]	$1/R$ median [μm^{-1}]	Digitization interval	Minimum crossings
NO	0.3	51.2746	51.1204–51.3168	2
NO	1.0	172.6645	171.4732–173.7360	2
IO	0.3	54.3497	54.2921–54.3824	2
IO	1.0	180.0192	179.9481–180.8174	3

In every case, the open comparator crosses the project threshold $\Delta\chi^2 = 4.605170185988092$ multiple times within the narrow digitization interval. Therefore none of the four points defines a unique pointwise replication candidate. The correct interpretation is *local numerical topology unresolved in the open comparator*; it is not evidence that the published contour is wrong or non-replicable.

5.2 Correction of an earlier topology attribution

An earlier version of the audit reported 39 threshold crossings for $N = 192$ and 17 for $N = 384$. Reviewer-style inspection identified a confound: the two truncations had been evaluated on different radius-grid densities. A matched-grid rerun withdrew that interpretation. On common grids, the crossing counts increased from 65/67 at 318 points to 199/203 at 635 points and 331/333 at 1269 points for $N = 192/384$, respectively. Neither truncation was internally stable under radius-grid refinement. At the finest common grid, however, the two $\Delta\chi^2$ curves were nearly identical pointwise (Pearson correlation 0.9999988; median absolute difference 0.0129). The earlier claim of KK-truncation-dependent micro-topology is therefore not retained.

6 Public Daya Bay control and correction of the calibration claim

The Daya Bay Collaboration has released the full 3158-day data set and states that the public material is sufficient to reproduce the standard oscillation-parameter measurement.^{7,8} The public software stack separates the model from the analysis scripts; this distinction matters for interpreting the earlier audit.

The public covariance-fit reference has $\chi^2 = 557.9755550235543$. Running the public full-fit route in the same frozen software environment reproduced that value as 557.9755550235410, an absolute difference of 1.33×10^{-11} . All 21 best-fit parameters were reproduced to a maximum absolute difference of 5.77×10^{-11} .

By contrast, the project’s earlier fixed-oscillation profile over 19 spectrum-shape parameters returned $\chi^2 = 574.1120550199523$, identical to the archived worker result but 16.1365 above the public full-fit reference. The correct conclusion is not that the Daya Bay public model failed calibration. Rather, the fixed-coordinate profiling procedure was computationally reproducible but not equivalent to the public full-fit procedure.

Earlier project-defined binary thresholds on absolute χ^2 , Spearman correlation, and $\Delta\Delta\chi^2$ summaries are retained only as historical audit diagnostics and are not used as scientific pass/fail criteria in this manuscript.

7 Reproducibility status and external continuous integration

The current Figure 1 replication is publicly available at the frozen repository revision 73728590833e. The repository contains the independent D29A Figure 1 calculation, the D29B conditioning-aware validator, an exact Python environment, and the one-command runner `reproduce_figure1.sh`.

The same frozen revision was executed independently on GitHub-hosted Ubuntu in workflow run 36403948683. The hosted runner reported Ubuntu 24.04.5 and Python 3.12.14.

The workflow reproduced the deliberately preserved D29A automatic raw-residual failure, all nine successful finite/infinite triangulation checks, all three $> 99\%$ unitarity checks, and the final D29B verdict `PASS_FIGURE1_COMPUTATIONAL_CORE_INDEPENDENT_REPLICATION`.

The CI verification step completed with `FIGURE1_REPLICATION_CI_PASS`.

The uploaded workflow artifact is ID 10962090038; GitHub reported SHA-256 9f0d1b782f9995f2c437944

The artifact contains the D29A and D29B result JSON files, the three root tables, and the independently generated Figure 1-style spectrum. This external run therefore tests the current manuscript-facing replicated subset rather than only historical D25–D27 outputs.

8 Limitations and claim boundaries

The successful result is deliberately narrow. The present work establishes an independent reproduction of the Figure 1 Brane–Dirac computational spectrum from the published equations. It does not reconstruct the target authors’ full numerical infrastructure, experimental likelihoods, or the published Figure 5 exclusion contours.

The Figure 5 comparator uses open experimental products and a related public implementation, not the target authors’ unavailable likelihood pipeline. Multiple local threshold crossings in that comparator therefore diagnose the open reconstruction, not the original article. The target paper’s experiment-level results are neither validated nor falsified here.

The conditioning correction after the initial D29A automatic failure is also explicitly bounded. The original raw-residual failure is retained in the audit trail. The later correction does not loosen that residual threshold; it demonstrates that raw residual magnitude is ill-conditioned as a root-coordinate accuracy metric near cotangent poles and validates the roots against the solver-coordinate tolerance and two high-precision calculations.

The reviewer-runnable Figure 1 workflow is now public and has passed a clean GitHub-hosted Ubuntu execution at the frozen revision given in Section 7. This establishes reproducibility of the manuscript-facing computation, while ordinary ReScience review still determines whether the demonstrated subset is sufficiently substantial for publication.

9 Conclusion

The Brane–Dirac spectrum underlying Figure 1 of de Giorgi, Pasari, and Turner was independently reproduced from the published equations. Two independent numerical routes agree at nine preregistered anchors to at worst 1.29×10^{-9} in mass, and the target article’s $> 99\%$ unitarity criterion is satisfied for all three published μ_1 choices. A

conditioning-aware audit over 3552 roots and two 80-digit checks resolves an initially failed raw-residual diagnostic without changing the physical calculation.

The replication does not extend to the selected Figure 5 exclusion points: the tested open comparator has multiple local threshold crossings inside each digitization interval, so a unique pointwise mapping to the published contour is not established. A separate public Daya Bay control reproduces the covariance-fit reference to numerical precision and shows that an earlier fixed-coordinate profiling route was non-equivalent rather than a failure of the public reference analysis.

The resulting claim is therefore bounded but positive: **Figure 1 is the successfully replicated computational subset; Figure 5 remains unresolved in the open reconstruction.** The [Re] classification is restricted to that Figure 1 subset and remains subject to the ordinary ReScience C editorial and reviewer standard.

Replication-status statement

This manuscript uses the [Re] prefix only for the independently reproduced Figure 1 Brane–Dirac spectrum subset. ReScience C’s current reviewer guidance defines a partial replication as a successful replication of a subset of the original results. The experiment-level Figure 5 exclusion contours remain unresolved in the open reconstruction reported here and are explicitly excluded from the successful-replication claim. We do not use $[\neg\text{Re}]$ because the present audit does not establish that the original exclusion analysis is non-replicable or mistaken.

AI-use disclosure

OpenAI ChatGPT was used as an assistive tool for code drafting, audit organisation, manuscript drafting, and consistency checking. Numerical claims in this article were accepted only when reproduced by executable scripts and traced to frozen result artifacts or primary sources. The author selected the research questions, executed the computational packages, reviewed the outputs, and remains responsible for the scientific claims and the manuscript.

Competing interests

The author declares no competing interests.

Code and data

Frozen public code revision: <https://github.com/yokken0907/led-neutrino-open-reproduction-audit/tree/73728590833e10c85eb2dd75bf844b825271b53d>. The manuscript-facing Figure 1 workflow was independently rerun on GitHub-hosted Ubuntu in workflow run 36403948683, which completed successfully and uploaded artifact 10962090038 (SHA-256 9f0d1b782f9995f2c4379446). Daya Bay public data authority: <https://doi.org/10.5281/zenodo.17587229>.

References

1. H. Davoudiasl, P. Langacker, and M. Perelstein. “Constraints on large extra dimensions from neutrino oscillation experiments.” In: *Physical Review D* 65 (2002), p. 105015. doi: 10.1103/PhysRevD.65.105015.
2. P. A. N. Machado, H. Nunokawa, and R. Zukanovich Funchal. “Testing for large extra dimensions with neutrino oscillations.” In: *Physical Review D* 84 (2011), p. 013003. doi: 10.1103/PhysRevD.84.013003.

3. D. V. Forero, C. Giunti, C. A. Ternes, and O. Tyagi. "Large extra dimensions and neutrino experiments." In: *Physical Review D* 106 (2022), p. 035027. doi: 10.1103/PhysRevD.106.035027. arXiv: 2207.02790 [hep-ph].
4. P. Eller, M. Ettengruber, and A. Zander. "Neutrino data analysis of extra-dimensional theories with massive bulk fields." In: *Physical Review D* 112 (2025), p. 055009. doi: 10.1103/11lm-96vy.
5. A. de Giorgi, D. Pasari, and J. Turner. "Do neutrinos dream in 5D? Towards a comprehensive extra-dimensional neutrino phenomenology." In: *Journal of High Energy Physics* 2026.5 (2026), p. 152. doi: 10.1007/JHEP05(2026)152. arXiv: 2512.02101 [hep-ph].
6. T. G. Elaçmaz, I. Martinez-Soler, and Y. F. Perez-Gonzalez. "Updated constraints on large extra dimensions from reactor antineutrino experiments." In: *Physical Review D* 114 (2026), p. 055014. doi: 10.1103/zp9f-27kk. arXiv: 2510.12900 [hep-ph].
7. Daya Bay Collaboration. "Precision Measurement of Reactor Antineutrino Oscillation at Kilometer-Scale Baselines by Daya Bay." In: *Physical Review Letters* 130 (2023), p. 161802. doi: 10.1103/PhysRevLett.130.161802.
8. Daya Bay Collaboration. *Full Data Release of the Daya Bay Reactor Neutrino Experiment*. Version 1.0.0. Zenodo, 2025. doi: 10.5281/zenodo.17587229.